Ch 5 S7 9-27

changing dependent variables

$x, y \to \xi, \eta$

~~du =~~ ~~check~~ check $J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} \neq 0 \pm \infty$

$$du = u_\xi \, d\xi + u_\eta \, d\eta$$

$$\frac{\partial u}{\partial x}\Big|_y = u_\xi \frac{\partial \xi}{\partial x}\Big|_y + u_\eta \frac{\partial \eta}{\partial x}\Big|_y$$

$$\frac{\partial u}{\partial y}\Big|_x = u_\xi \frac{\partial \xi}{\partial y}\Big|_x + u_\eta \frac{\partial \eta}{\partial y}\Big|_x$$

Not $= \begin{vmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{vmatrix}$ not = but gives same result.

see below

$$= \begin{vmatrix} x_\xi & y_\xi \\ x_\eta & y_\eta \end{vmatrix}$$

$\det A = \det A^T$? yes

<u>Note</u> If A^{-1} is the inv. of matrix A

Then $\det(A^{-1})\det(A) = \det(A^{-1}A) = \det(I) = 1$

$\Rightarrow$ If $\det A \neq 0, \pm\infty$ Then $\det(A^{-1}) \neq 0, \pm\infty$

Question: is $\begin{bmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{bmatrix}^{-1} = \begin{bmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{bmatrix}$? mult. + see if get I

$$\begin{bmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{bmatrix}\begin{bmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{bmatrix} = \begin{bmatrix} \frac{\partial \xi}{\partial x}\frac{\partial x}{\partial \xi} + \frac{\partial \xi}{\partial y}\frac{\partial y}{\partial \xi} & \frac{\partial \xi}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial \xi}{\partial y}\frac{\partial y}{\partial \eta} \\ \frac{\partial \eta}{\partial x}\frac{\partial x}{\partial \xi} + \frac{\partial \eta}{\partial y}\frac{\partial y}{\partial \xi} & \frac{\partial \eta}{\partial x}\frac{\partial x}{\partial \eta} + \frac{\partial \eta}{\partial y}\frac{\partial y}{\partial \eta} \end{bmatrix} = I$$

$$= \begin{bmatrix} \frac{\partial \xi}{\partial \xi} & \frac{\partial \xi}{\partial \eta} \\ \frac{\partial \eta}{\partial \xi} & \frac{\partial \eta}{\partial \eta} \end{bmatrix}$$

Use $d\xi = \frac{\partial \xi}{\partial x}dx + \frac{\partial \xi}{\partial y}dy$

$$\frac{\partial \xi}{\partial \xi}\Big|_\eta = \frac{\partial \xi}{\partial x}\frac{\partial x}{\partial \xi}\Big|_\eta + \frac{\partial \xi}{\partial y}\frac{\partial y}{\partial \xi}\Big|_\eta = 1$$

etc

2nd-order?

u_{xx} + I want $u_{\xi\xi}, u_{\eta\eta}$

$$\frac{\partial}{\partial x}\Big[\underbrace{u_x}_{w}\Big]$$

$$dw = \frac{\partial w}{\partial \xi} d\xi + \frac{\partial w}{\partial \eta} d\eta$$

$$\frac{\partial}{\partial x}\left[du_x = \frac{\partial u_x}{\partial \xi} d\xi + \frac{\partial u_x}{\partial \eta} d\eta\right]$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u_x}{\partial x}\Big|_y = \frac{\partial u_x}{\partial \xi}\underbrace{\frac{\partial \xi}{\partial x}\Big|_y}_{\text{easy}} + \frac{\partial u_x}{\partial \eta}\underbrace{\frac{\partial \eta}{\partial x}\Big|_y}_{\text{easy}}$$

$$\frac{\partial u_x}{\partial \xi} = \frac{\partial}{\partial \xi}\left[\frac{\partial u}{\partial x}\right] = \frac{\partial}{\partial \xi}\left[u_\xi \underbrace{\frac{\partial \xi}{\partial x}\Big|_y}_{\text{know}} + u_\eta \underbrace{\frac{\partial \eta}{\partial x}\Big|_y}_{\text{know}}\right]$$

$$= \frac{\partial^2 u}{\partial \xi^2}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \xi}\underbrace{\frac{\partial}{\partial \xi}\left(\frac{\partial \xi}{\partial x}\Big|_y\right)}_{\text{careful!}} + \frac{\partial^2 u}{\partial \xi \partial \eta}\frac{\partial \eta}{\partial x} + \frac{\partial u}{\partial \eta}\underbrace{\frac{\partial}{\partial \xi}\left(\frac{\partial \eta}{\partial x}\Big|_y\right)}_{\text{careful!}}$$

careful! must convert $\frac{\partial \xi}{\partial x}$ as a fn of ξ, η

careful!

$$\frac{\partial}{\partial \xi}\Big|_\eta\left(\frac{\partial \xi}{\partial x}\Big|_y\right) = \frac{\partial}{\partial x}\left(\frac{\partial \xi}{\partial \xi}\right) = 0$$

NO! NO! NO!

can only switch when

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{or} \quad \frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{\partial^2 u}{\partial \eta \partial \xi}$$

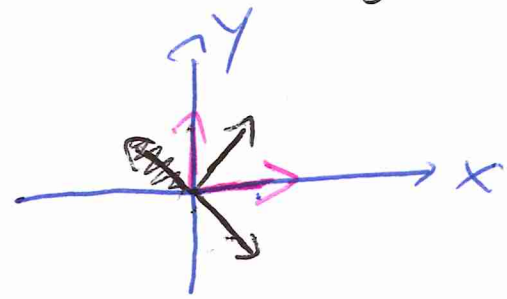
ex) $u_{xx} - u_{yy} = 0$ Type?

$a=1 \quad b=0 \quad c=-1$

$b^2 - 4ac = 0 - (4)(1)(-1) = 4 > 0$ hyperbolic

$\xi = x+y \quad \eta = x-y$

~~rotation by 45°~~



ex) $x=1, y=0$
$\Rightarrow (\xi,\eta) = (1,1)$

$x=0, y=1$
$(\xi,\eta) = (1,-1)$

$du = u_x dx + u_y dy$
$= u_\xi d\xi + u_\eta d\eta$

$\left.\frac{\partial u}{\partial x}\right|_y = u_\xi \left.\frac{\partial \xi}{\partial x}\right|_y + u_\eta \left.\frac{\partial \eta}{\partial x}\right|_y$
$= u_\xi(1) + u_\eta(1)$ (*)

$\left.\frac{\partial u}{\partial y}\right|_x = u_\xi \left.\frac{\partial \xi}{\partial y}\right|_x + u_\eta \left.\frac{\partial \eta}{\partial y}\right|_x$
$= u_\xi(1) + u_\eta(-1)$ (**)

Think hmmm...

$J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 \neq 0$

See next 2 pp for an aside

Aside Linear transformations (from linear algebra) rotations

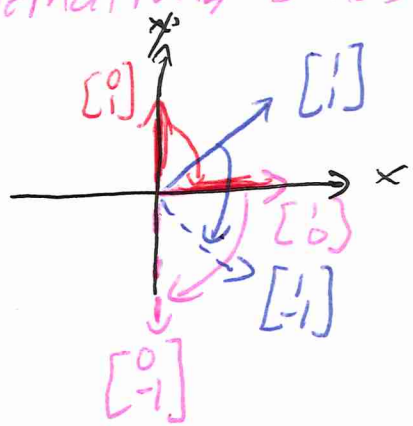
To rotate a vector by an angle θ clockwise $\quad A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

check $\begin{bmatrix} \cos\frac{\pi}{2} & \sin\frac{\pi}{2} \\ -\sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

multiply by $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} x \\ y \end{bmatrix}$

This transformations sends $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \to \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \to \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\xi = 0x + 1y = y$
$\eta = -x + 0y = -x$



Note that $J = \begin{vmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 1$

Since $J = 1$, the volume element does not change

rotate by 45° clockwise

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{\sqrt{2}}{2} \begin{bmatrix} x + y \\ -x + y \end{bmatrix}$$

$$J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \left(\frac{\sqrt{2}}{2}\right)^2 \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 1$$

rotate by 45° CCW

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} \cos(-45^\circ) & \sin(-45^\circ) \\ -\sin(-45^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{\sqrt{2}}{2} \begin{bmatrix} x - y \\ x + y \end{bmatrix}$$

$$J = \left(\frac{\sqrt{2}}{2}\right)^2 \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 1$$

My proposed rotation

$$\xi = x+y \qquad \eta = x-y$$

corresponds to

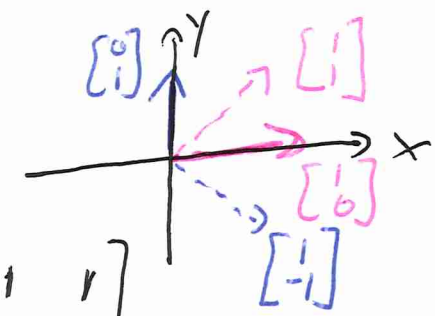
$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

cannot be a rotation because $\cos\theta = \cos(-\theta)$ (trouble)

also $J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$ → indicates volume change

↑ indicates a flip of an axis

<u>check</u> $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix} \to \begin{bmatrix} 1 \\ -1 \end{bmatrix}$



I think this is a rotation by -45° followed by a flip about $y=x$.

$$\underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{T(e_1)\ \ T(e_2)} \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_{\text{rotate by } -45^\circ} \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ yep.}$$

I believe this coord. transformation still transforms $u_{xx} - u_{yy} = 0$ to $u_{\xi\eta} = 0$

$$d\,u_x = \frac{\partial u_x}{\partial \xi}\,d\xi + \frac{\partial u_x}{\partial \eta}\,d\eta$$

$$\left.\frac{\partial u_x}{\partial x}\right|_y = \frac{\partial u_x}{\partial \xi}\left.\frac{\partial \xi}{\partial x}\right|_y + \frac{\partial u_x}{\partial \eta}\left.\frac{\partial \eta}{\partial x}\right|_y$$

$$= \frac{\partial u_x}{\partial \xi}(1) + \frac{\partial u_x}{\partial \eta}(1)$$ use (*) ~~+ u(**)~~

$$= \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}\right) + \frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}\right)$$

$$= u_{\xi\xi} + u_{\eta\xi} + u_{\xi\eta} + u_{\eta\eta}$$

$$u_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$$

first second

$$u_{xx} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}$$

Do same for u_{yy}

$$u_{xx} - u_{yy} = 4u_{\xi\eta} = 0$$

$u_{xx} - u_{yy} = 0$

or $u_{xy} = 0$

$a = 0 \quad b = 1 \quad c = 0$

$b^2 - 4ac = 1 > 0$ hyperbolic

§5.9 Principle of Superposition.

Let $P(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha D^\alpha$ (linear operator)

(coeff. cannot be a f'n of dep. variable)

$$P(c_1 u_1 + c_2 u_2) = c_1 P u_1 + c_2 P u_2$$

ex) $P = x \frac{\partial}{\partial x} + x^2 \frac{\partial}{\partial y}$

$$\text{LHS} = x \frac{\partial}{\partial x}(c_1 u_1 + c_2 u_2) + x^2 \frac{\partial}{\partial y}(c_1 u_1 + c_2 u_2)$$

$$= x c_1 \frac{\partial u_1}{\partial x} + x c_2 \frac{\partial u_2}{\partial x} + x^2 c_1 \frac{\partial u_1}{\partial y} + x^2 c_2 \frac{\partial u_2}{\partial y}$$

$$= c_1 \left(x \frac{\partial u_1}{\partial x} + x^2 \frac{\partial u_1}{\partial y} \right) + c_2 \left(x \frac{\partial u_2}{\partial x} + x^2 \frac{\partial u_2}{\partial y} \right)$$

$$= \text{RHS}$$

other examples of linear operators

① integration

$$\int (c_1 f_1 + c_2 f_2)\, dx \overset{?}{=} c_1 \int f_1 + c_2 \int f_2$$ yes linear

② Matrix Mult. $A(c_1 \underline{u}_1 + c_2 \underline{u}_2)$

$= c_1 A \underline{u}_1 + c_2 A \underline{u}_2$

This is powerful!

ex) $\{u_j\}_{j=1}^{\infty}$ are all sol'ns to a linear PDE

$\Rightarrow \sum_{j=1}^{\infty} c_j u_j$ is also a sol'n, provided we can interchange P & $\sum_{j=1}^{\infty}$

ex) $u(x,\lambda)$ is a sol'n for all λ $\quad Pu=0$

Then $u(x) = \int_I g(\lambda)\, u(x,\lambda)\, d\lambda$ is also a sol'n, provided we can interchange $\int$ & P